



JOINT QUANTIZATION OF MEASURES FOR ML

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INTRODUCCIÓN

In machine learning, it is common to approximate complex structures through low-dimensional discrete representations.

Classical example: k -means clustering.

Given a dataset $\{x_i\}_{i=1}^n$, we seek centers $a_1, \dots, a_k$ that minimize

$$\frac{1}{n} \sum_{i=1}^n \min_{1 \leq j \leq k} d(x_i, a_j)^2.$$

Central question:

Can this paradigm be extended to the space of *probability measures*?

PRELIMINARIES

Wasserstein Distance

Let (E, d) be a metric space and let $\mu, \nu \in \mathcal{W}_p(E)$. The *Wasserstein distance of order p* between μ and ν is defined as

$$W_p(\mu, \nu) = \left(\inf_{\pi \in \Pi(\mu, \nu)} \int_{E \times E} d(x, y)^p d\pi(x, y) \right)^{\frac{1}{p}}$$

where $\Pi(\mu, \nu)$ denotes the set of all couplings of μ and ν , i.e., all probability measures on $E \times E$ with marginals μ and ν .

Equivalently, $X \sim \mu, Y \sim \nu$

$$W_p(\mu, \nu) = \inf \left\{ (\mathbb{E}[d(X, Y)^p])^{1/p} \right\}.$$

Wasserstein Barycenter

Let (E, d) be a metric space. Given probability measures $\mu_1, \dots, \mu_n \in \mathcal{W}_p(E)$ and weights $\lambda_1, \dots, \lambda_n \geq 0$ such that $\sum_{i=1}^n \lambda_i = 1$, a *Wasserstein barycenter of order p* is any measure $\nu^* \in \mathcal{W}_p(E)$ satisfying

$$\nu^* \in \operatorname{argmin}_{\nu \in \mathcal{W}_p(E)} \sum_{i=1}^n \lambda_i W_p^p(\nu, \mu_i). \quad (1)$$

QUANTIZATION

Functional Formulation: Let (E, d) be a metric space and let X be a random variable with distribution μ . For $k \in \mathbb{N}$, define

$$\mathcal{F}_k = \{f : E \rightarrow E \text{ Borel measurable} : |f(E)| \leq k\}.$$

Elements of $\mathcal{F}_k$ are called *k -quantizers*. For $f \in \mathcal{F}_k$, the random variable $f(X)$ represents a quantized version of X .

The *k -th quantization error of order p* is defined by $\inf_{f \in \mathcal{F}_k} \mathbb{E}[d(X, f(X))^p]$. A quantizer achieving this minimum is called *k -optimal*.

k -means-type formulation: For fixed k , the problem is equivalent to

$$\inf_{\substack{\alpha \in E \\ |\alpha| \leq k}} \mathbb{E} d(X, \alpha)^p = \inf_{f \in \mathcal{F}_k} \mathbb{E}[d(X, f(X))^p],$$

where $d(x, \alpha) := \min_{a \in \alpha} d(x, a)$.

Measure-Based (Wasserstein) Formulation: The problem can then be rewritten as

$$V(k, \mu) = \inf_{\nu \in \mathcal{M}_k} W_p^p(\mu, \nu).$$

We say that $\mu_k^* \in \mathcal{M}_k$ is an *optimal quantizer* if $W_p(\mu, \mu_k^*) = \inf_{\nu \in \mathcal{M}_k} W_p(\mu, \nu)$.

where $\mathcal{M}_k$ denotes the set

$$\left\{ \nu = \sum_{i=1}^k \lambda_i \delta_{x_i} : x_i \in E, \lambda_i \geq 0, \sum_{i=1}^k \lambda_i = 1 \right\}.$$

Theorem 4.12 in *Foundations of Quantization for Probability Distributions* establishes the **existence** of an optimal quantization set.

Lemma 2

Let $\mu \in \mathcal{W}_p(E)$. Then,

$$\inf_{\mu_k \in \mathcal{M}_k} W_p^p(\mu_k, \mu) = \inf_{\substack{\alpha \subset E \\ |\alpha| \leq k}} \mathbb{E} d(X, \alpha)^p = \inf_{f \in \mathcal{F}_k} \mathbb{E}[d(X, f(X))^p].$$

Theorem 4

Every k -optimal quantizer f for a measure $\mu \in \mathcal{W}_p(E)$ induces an optimal discrete measure $\nu_k^* \in \mathcal{M}_k$ given by

$$\nu_k^* := \mu \circ f^{-1}. \quad (2)$$

Proposition

Let $\mu_k^* \in \mathcal{M}_k$ satisfy $W_p(\mu_k^*, \mu) = V(k, \mu)$. Denote by $\alpha = \{a_1, \dots, a_k\}$ the support of μ_k^* and consider the associated Voronoi partition $\{A_1, \dots, A_k\}$ of E . Then the map

$$g = \sum_{i=1}^k a_i \mathbf{1}_{A_i}$$

is a k -optimal quantizer. Moreover, α solves the corresponding k -means problem.

CONVERGENCE

Theorem 5

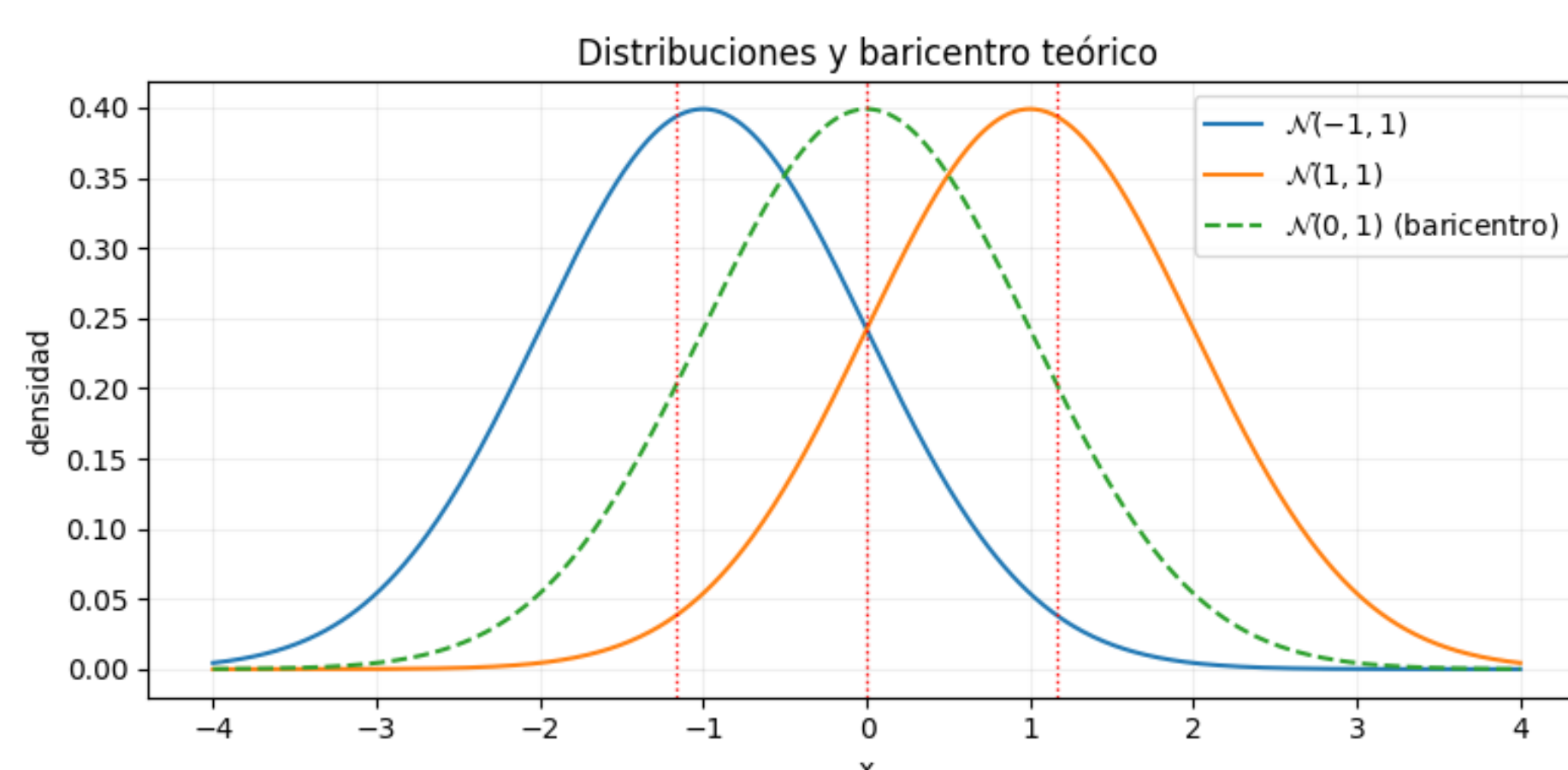
Let E be a complete and separable metric space, and let $\mu_k^* \in \mathcal{M}_k$ be a k -optimal quantizer of $\mu \in \mathcal{W}_p(E)$. Then

$$\ell(\mu_k^*; \mu) \longrightarrow 0 \quad \text{as } k \rightarrow \infty,$$

that is, $\lim_{k \rightarrow \infty} W_p(\mu, \mu_k^*) = 0$.

Probability measures with finite support are dense in $W_p(E)$.

EXAMPLE



JOINT QUANTIZATION

The *k -th quantization error of order p* for two measures $\mu_1, \mu_2 \in \mathcal{W}_p(E)$ is defined as

$$V(k; \mu_1, \mu_2) = \inf_{\mu_k \in \mathcal{M}_k} \left(\frac{1}{2} W_p^p(\mu_k, \mu_1) + \frac{1}{2} W_p^p(\mu_k, \mu_2) \right)$$

We say that $\mu_k^* \in \mathcal{M}_k$ is an *optimal joint quantizer* for μ_1 and μ_2 if

$$\ell(\mu_k^*; \mu_1, \mu_2) = \inf_{\mu_k \in \mathcal{M}_k} \ell(\mu_k; \mu_1, \mu_2),$$

where

$$\ell(\nu; \mu_1, \mu_2) := \frac{1}{2} W_p^p(\nu, \mu_1) + \frac{1}{2} W_p^p(\nu, \mu_2).$$

Relation to the barycenter. The Wasserstein barycenter μ^* of μ_1 and μ_2 is defined as

$$\mu^* = \operatorname{argmin}_{\nu} \left(\frac{1}{2} W_p^p(\nu, \mu_1) + \frac{1}{2} W_p^p(\nu, \mu_2) \right).$$

Question. Is quantizing the barycenter equivalent to jointly quantizing μ_1 and μ_2 ?

For fixed k , the two problems do not coincide.

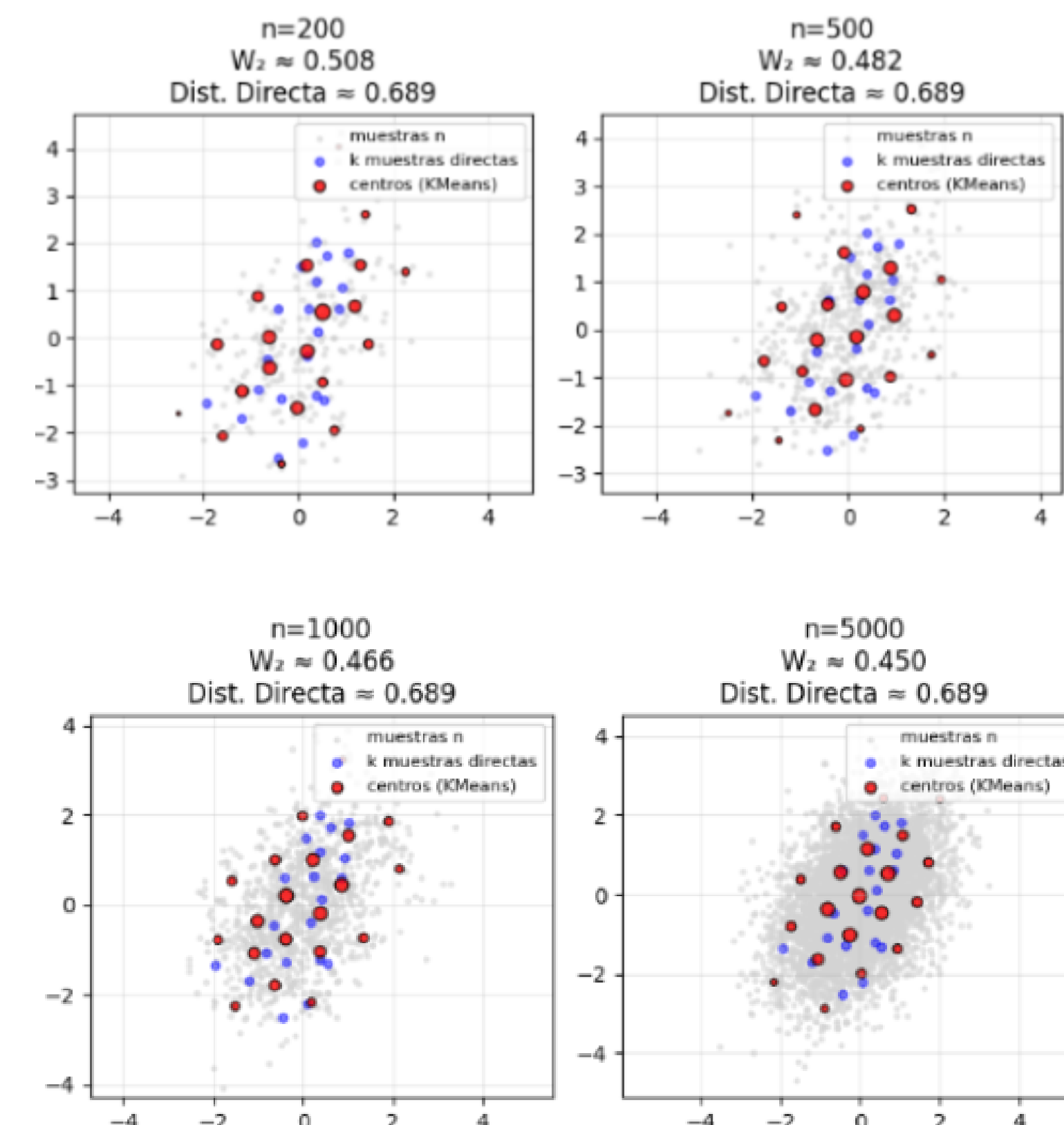
Theorem 6 – Barycenter Approximation

Let μ_1 and μ_2 be probability measures, and let μ^* denote their (unique) Wasserstein barycenter with respect to W_p . Let Q_k^* be an optimal joint quantizer of μ_1 and μ_2 supported on at most k points. Then

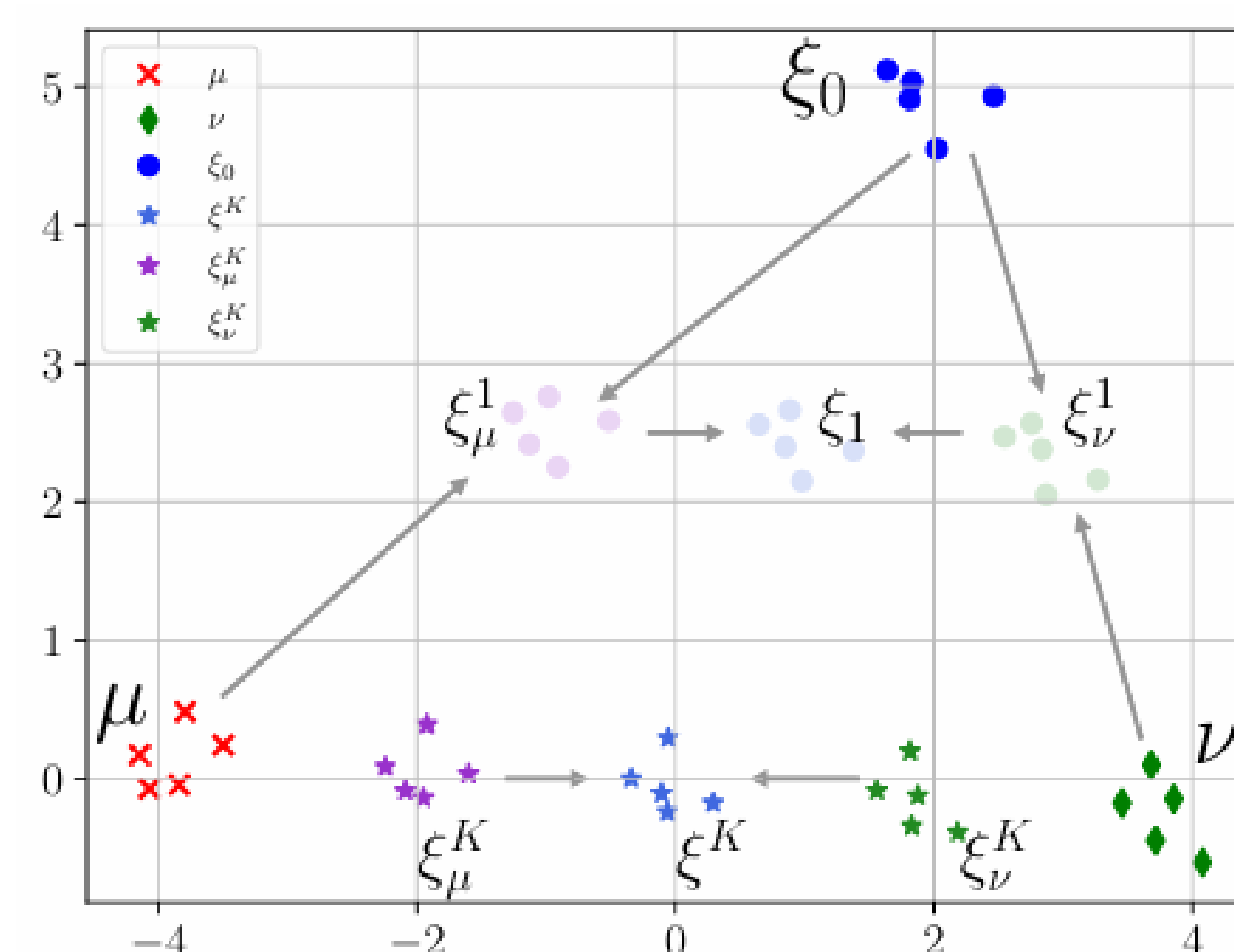
$$Q_k^* \xrightarrow[k \rightarrow \infty]{D} \mu^*.$$

APPLICATIONS IN ML

1. Complexity reduction: Quantization enables the representation of complex probability distributions using a small number of support points while preserving their underlying geometric structure.



2. Federated Learning:



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